The Terminal Grid Duel

Drayan Mathematical Series

Problem

Ardra and Bumblebee play a game. Initially, they write the pair $(1012, 1012)$ on the board. They take turns alternately, with Ardra starting first. In each move, a player replaces the current pair (a, b) with one of the following, provided both entries remain nonnegative:

$$(a, b) \to (a-2, b+1), \quad (a, b) \to (a+1, b-2), \quad (a, b) \to (a-1, b).$$

The game ends when no legal move is possible.
Ardra wins if the game ends at $(0, 0)$, and Bumblebee wins if the game ends at $(0, 1)$.
Determine which player has a winning strategy.